

LEVEL 2 — E-COMMERCE SALES DATA ANALYSIS

Cognevance Technologies

Data Analysis with Python — Project Report

1. Project Overview

The **E-Commerce Sales Data Analysis** project focuses on analyzing an e-commerce sales dataset to understand **product performance, customer behavior, revenue trends, top-selling products, and peak sales periods**.

The project was developed using **Python**, with **Pandas and NumPy** for data cleaning and analysis and **Matplotlib** for creating visualizations.

The complete workflow includes data preprocessing, sales analysis, customer analysis, revenue trend analysis, identification of top-performing products and sales periods, visualization, and generation of business insights and recommendations.

This project follows the Level 2 — Intermediate requirements provided by Cognevance Technologies.

2. Project Objectives

The main objectives of this project are:

- To collect and analyze an e-commerce sales dataset.
- To clean and preprocess the data using Pandas and NumPy.
- To analyze product sales performance.
- To understand customer behavior.
- To analyze revenue trends over time.
- To identify top-selling products.
- To identify peak sales periods.
- To create meaningful dashboards/visualizations using Python libraries.
- To generate actionable business insights and recommendations.
- To prepare a detailed analysis report.

These objectives correspond to the Level 2 project requirements.

3. Dataset Overview

Attribute	Details
Dataset Name	Global E-Commerce Sales Dataset
File Name	global_ecommerce_sales.csv
Number of Records	2,000
Number of Columns	15
Time Period	2023–2025
Programming Language	Python
Environment	Google Colab
Main Libraries	Pandas, NumPy, Matplotlib

Dataset Columns

The dataset contains the following fields:

- Order_ID
- Order_Date
- Customer_Name
- Customer_Segment
- Country
- Region
- Product_Category
- Product_Name
- Quantity
- Unit_Price
- Discount_Percent
- Total_Sales
- Shipping_Cost
- Profit
- Payment_Method

These fields provide information about orders, customers, products, sales, shipping, profit, and payment methods.

4. Data Cleaning and Preprocessing

The dataset was loaded and processed using **Pandas** and **NumPy**.

The following data-quality checks were performed.

4.1 Data Loading

The `global_ecommerce_sales.csv` dataset was loaded into a Pandas DataFrame for analysis.

4.2 Dataset Structure

The dataset contains:

- **2,000 rows**
- **15 columns**

The structure and data types of the columns were checked before performing analysis.

4.3 Missing Values

All columns were checked for missing values.

Result: No missing values were found in the dataset.

4.4 Duplicate Records

The dataset was checked for duplicate rows.

Result: No duplicate records were found.

4.5 Date Conversion

The `Order_Date` column was converted into datetime format to support time-based revenue analysis and monthly grouping.

4.6 Incorrect Record Validation

The following fields were checked for invalid values:

- Quantity
- Unit Price
- Total Sales
- Shipping Cost

- Discount Percentage

The checks confirmed that:

- Quantity did not contain non-positive values.
- Unit Price did not contain non-positive values.
- Total Sales did not contain non-positive values.
- Shipping Cost did not contain negative values.
- Discount Percentage remained within the valid 0–100% range.

Final Data Quality Result

No missing values, duplicate rows, or identified invalid records were found. Therefore, **no major data-cleaning modifications were required** after validation.

5. Product Sales Analysis

Product sales were analyzed by grouping the data according to **Product Category** and calculating total sales.

Total Sales by Product Category

Product Category	Total Sales
Furniture	256,274.68
Technology	139,518.22
Clothing & Accessories	69,375.63
Office Supplies	19,390.81

Observation

Furniture generated the highest total sales among all product categories, with total sales of **256,274.68**.

Office Supplies generated the lowest total sales at **19,390.81**.

This indicates that Furniture was the strongest-performing product category in the dataset.

6. Customer Behavior Analysis

Customer behavior was analyzed using the **Customer_Segment** field.

Total sales were calculated for each customer segment.

Sales by Customer Segment

Customer Segment	Total Sales
Consumer	256,287.74
Corporate	146,050.39
Home Office	82,221.21

Observation

The **Consumer** segment generated the highest total sales, with **256,287.74**.

The **Home Office** segment generated the lowest total sales, with **82,221.21**.

This shows that Consumer customers represent the strongest sales segment in the analyzed dataset.

7. Revenue Trend Analysis

Revenue trends were analyzed by grouping total sales according to the month of the order date.

Monthly revenue was calculated for the period **2023–2025**.

The analysis was used to identify variations in sales across different months and to determine the strongest sales periods.

Highest Revenue Periods

Period	Total Sales
June 2025	18,068.09
October 2023	17,794.60
May 2025	17,621.80
October 2024	17,426.05

February 16,943.67
2023

Observation

June 2025 recorded the highest monthly sales in the dataset, with total sales of **18,068.09**.
The monthly analysis shows that sales varied considerably across different periods.

8. Top-Selling Products

The total sales generated by each product were calculated to identify the highest-performing products.

Top-Selling Products

Rank	Product	Total Sales
1	Standing Desk Converter	46,614.22
2	Ergonomic Office Chair	45,405.15
3	Corner L-Shaped Desk	41,070.48
4	Mesh Back Task Chair	38,179.77
5	Wireless Bluetooth Headphones	28,258.70

Observation

The **Standing Desk Converter** was the highest-selling product, generating **46,614.22** in total sales.
Several of the highest-performing products were office furniture products, which is consistent with the strong performance of the Furniture category.

9. Peak Sales Periods

Monthly sales were sorted in descending order to identify periods with the highest revenue.

The strongest sales period was:

June 2025 — 18,068.09

Other high-performing periods included:

- October 2023
- May 2025
- October 2024
- February 2023

Identifying these periods can help businesses understand when sales performance was strongest and support future sales and inventory planning.

10. Data Visualizations

The project includes visualizations created using **Matplotlib**.

The following charts were prepared:

10.1 Total Sales by Product Category

This visualization compares total sales generated by each product category.

10.2 Sales by Customer Segment

This chart compares sales performance across Consumer, Corporate, and Home Office customer segments.

10.3 Monthly Revenue Trend

This line chart shows how total sales changed across the months from 2023 to 2025.

10.4 Top 10 Selling Products

This visualization displays the ten products with the highest total sales.

10.5 Top 10 Peak Sales Periods

This chart highlights the ten months with the highest sales.

All visualizations were created using Python and saved as image files for the project repository.

11. Key Business Insights

The analysis produced the following key insights:

1. **Furniture is the highest-performing product category**, generating total sales of 256,274.68.
 2. **Office Supplies is the lowest-performing product category**, with total sales of 19,390.81.
 3. The **Consumer customer segment generates the highest sales**, with total sales of 256,287.74.
 4. The **Home Office segment generates the lowest sales**, with total sales of 82,221.21.
 5. The **Standing Desk Converter is the top-selling product**, generating 46,614.22 in sales.
 6. **June 2025 recorded the highest monthly sales**, reaching 18,068.09.
 7. The sales analysis indicates that office furniture products have a strong contribution to overall sales performance.
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12. Business Recommendations

Based on the analysis, the following recommendations can be made:

1. Focus on High-Performing Furniture Products

The business should maintain sufficient inventory and promotional focus on high-performing Furniture products.

2. Improve Office Supplies Sales

Targeted promotions, product bundles, or discounts can be considered to improve the performance of Office Supplies.

3. Target Consumer Customers

Since Consumers generate the highest sales, marketing campaigns and offers can be designed specifically for this customer segment.

4. Increase Home Office Sales

The business can explore targeted offers and product recommendations to increase purchases from Home Office customers.

5. Maintain Stock of Top-Selling Products

Products such as the Standing Desk Converter and Ergonomic Office Chair should have sufficient inventory to avoid missed sales opportunities.

6. Plan Around Strong Sales Periods

Historical peak periods can be considered when planning inventory, promotions, and marketing campaigns.

13. Project Workflow

The overall workflow followed in this project was:

E-Commerce Dataset Collection

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Data Loading using Pandas

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Data Cleaning & Preprocessing

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Missing Value & Duplicate Checks

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Data Validation

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Product Sales Analysis

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Customer Behavior Analysis

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Revenue Trend Analysis

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Top Product Identification

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Peak Sales Period Identification

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Data Visualization using Matplotlib

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Business Insights

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Business Recommendations

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Final Report

14. Tools and Technologies Used

Tool / Technology	Purpose
Python	Data analysis and programming
Pandas	Data loading, cleaning, grouping and analysis
NumPy	Numerical/data processing support
Matplotlib	Data visualization
Google Colab	Development environment
CSV	Dataset format
GitHub	Project repository and documentation

15. Project Deliverables

The completed project contains:

- Python analysis notebook
- E-commerce sales dataset
- Data-cleaning and analysis code
- Sales visualizations
- Business insights
- Business recommendations
- Detailed project report
- GitHub repository

These deliverables satisfy the expected outputs specified for the Level 2 E-Commerce Sales Data Analysis project.

16. Conclusion

The **E-Commerce Sales Data Analysis** project successfully demonstrates an end-to-end data analysis workflow using Python.

The dataset was loaded, validated, and analyzed using **Pandas and NumPy**, while **Matplotlib** was used to create meaningful visualizations.

The analysis provided insights into **product performance, customer segments, revenue trends, top-selling products, and peak sales periods**.

The results showed that Furniture was the strongest product category, Consumer customers generated the highest sales, the Standing Desk Converter was the top-selling product, and June 2025 was the highest-performing sales month.

Based on these findings, practical business recommendations were developed around inventory management, customer targeting, product promotion, and sales-period planning.

Overall, the project fulfills the requirements of **Level 2 — E-Commerce Sales Data Analysis** and demonstrates practical skills in **data cleaning, preprocessing, exploratory analysis, visualization, business insight generation, and analytical reporting**.

Project Information

Project: E-Commerce Sales Data Analysis

Level: Level 2 — Intermediate

Organization: Cognevance Technologies

Domain: Data Analysis

Language: Python

Environment: Google Colab

Libraries: Pandas, NumPy, Matplotlib

Dataset: `global_ecommerce_sales.csv`

Records: 2,000

Columns: 15

Data Period: 2023–2025